

Arushi Jha

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MS Data Science graduate with 2+ years of experience building AI/ML systems, data pipelines, and agentic solutions across fintech and enterprise domains. Skilled in Python, SQL, LLMs, RAG pipelines, and cloud-native architectures with a track record of deploying end-to-end ML systems that translate complex data into measurable business impact.

Education

University of Colorado Boulder, USA | MS in Data Science | GPA: 3.9/4.0 Aug 2024 – May 2026
VIT, India | Bachelor of Technology in Computer Science | GPA: 3.7/4.0 Aug 2018 – May 2022

Technical Skills

Programming: Python, SQL, R, Java
AI & ML: LangChain, LangGraph, RAG, Multi-Agent Systems, OpenAI GPT-4o, Scikit-learn, XGBoost, FAISS, Sentence-Transformers, SMOTE, NLP, Predictive Modeling
Data & Visualization: Pandas, NumPy, Tableau, Power BI, Matplotlib, Seaborn, Streamlit, Pinecone, Feature Engineering
Cloud & Databases: PostgreSQL, GCP (Cloud Run, Pub/Sub, Cloud SQL, Memorystore), Snowflake, Databricks, Azure, Redis, Docker
Tools & Platforms: Git, FastAPI, LangSmith, RAGAs, Jenkins, Jira, REST APIs, Pytest

Experience

Data Science Intern, Parlay Finance Jan 2026 – Apr 2026

- Designed a weighted scoring engine encoding **8** SBA domain parameters (NAICS, DSCR, credit, equity) across **341** production loan inquiries, programmatically generating ground-truth labels that enabled downstream ML training with **99%** F1 score.
- Applied SMOTE to resolve a **33:1** class imbalance, boosting XGBoost F1 from **78%** to **99%** and enabling reliable binary classification of loan eligibility across **50+** engineered features.
- Delivered a ranked loan product recommendation engine matching **100%** of eligible inquiries to qualifying SBA products at an average confidence score of **95%**, reducing manual lender review time and improving decision accuracy.

Software Engineer, NielsenIQ Feb 2022 – Jun 2024

- Built **4** ingestion pipelines centralizing processed data across **3** teams into Postgres, Snowflake, Databricks, and Azure Blob Storage, improving cross-functional data accessibility and reducing redundant data retrieval.
- Reduced data validation errors by **20%** across **100+** cross-mapped features and cut Spark ETL debug time by **95%** by building Angular and REST API monitoring dashboards with real-time pipeline visibility.
- Developed **2** order management services using Java and Spring Boot, optimizing deployments and end-to-end testing by **40%** through Jenkins, SonarQube, and Azure VMSS DevOps automation.

Data Science Student Assistant, University of Colorado Boulder Apr 2025 – May 2026

- Analyzed internship and job outcome data for **150+** students using Python and Excel, delivering Tableau dashboards that surfaced actionable trends for nontechnical stakeholders in program decision-making.
- Managed communications for **300+** students, led team meetings and event planning, improving response time by **30%** and enhancing program engagement and operational efficiency.

Projects

SkillRadar – Multi-Agent GenAI Skill Intelligence System GitHub

- Architected a **4-agent** LangGraph DAG (JD, RAG, Trend, Synthesis) integrating live job postings via JSearch API, arXiv research papers, and Hacker News signals to forecast role-specific skill demand **6–12 months** ahead.
- Engineered a Pinecone vector knowledge base (~**450** chunks, 1536-dim cosine index) ingested from **10** RSS feeds across major AI/ML platforms, powering an OpenAI-embedded RAG pipeline for semantic community signal retrieval.
- Designed the Skill Velocity Score (SVS) – a composite metric blending job demand (**50%**), community signals (**30%**), and research trends (**20%**) – deployed via FastAPI and Streamlit Cloud with LangSmith observability and an **80-test** pytest suite.

InsightLens – NLP Review Intelligence Pipeline GitHub

- Processed **568K** Amazon reviews through a Sentence-Transformer (MiniLM-L6-v2) embedding pipeline, achieving **80%** Precision@10 and **0.75** nDCG@10 via FAISS IndexFlatIP semantic search across **200K** L2-normalized vectors.
- Applied PCA (384→50 dims, 95% variance retained) and KMeans ($k=50$) to uncover **50** topic clusters, extracting TF-IDF driven strengths, pain points, and extractive summaries with a **90%** actionability and **95%** coverage score.

PigMint Finance – Cloud-Native Microservices Fintech Platform GitHub

- Deployed **3** Flask microservices to GCP Cloud Run with Pub/Sub async event processing, returning sub-second API responses by fully decoupling transaction ingestion from downstream savings computation.
- Implemented Redis cache-aside pattern (Memorystore) eliminating repeated Cloud SQL queries per transaction, reducing database load by **40%** and enabling scalable round-up detection across **7** normalized PostgreSQL tables.